

Sensitivity Workflow

Introduction

This presentation summarizes the workflow and outputs of the Sensitivity analysis for the SOLAR project.

- This deck is auto-generated from workflow outputs.
- This deck is scoped to the **Truncated** folder for the **SolarEnergy** reconstruction algorithm.

Config aliases:

- hd_1x2x6_centralAPA: HD Central
- hd_1x2x6_lateralAPA: HD Lateral
- vd_1x8x14_3view_30deg_nominal: VD Top
- vd_1x8x14_3view_30deg_shielded: VD Bottom Shielded

Workflow

- Orchestrator: `src/analysis/10SensitivityAnalysis.py`
- Step 1 (Background template): `src/analysis/14SensitivityBackgroundTemplate.py`
- Step 2 (Signal template): `src/analysis/14SensitivitySignalTemplate.py`
- Step 3 (Grid fit scan and best-cut storage): `src/analysis/14Sensitivity.py`
- Step 4 (Contour rendering): `src/analysis/14SensitivityContourPlot.py`

Workflow Outputs

- Main contour grids (sin12/sin13): `images/analysis/sensitivity`
- Signal/background templates (figures): `images/analysis/sensitivity/templates`
- Grid-scan data products (PKL): `data/analysis/sensitivity`
- Remote workflow outputs (PNFS):
`/pnfs/ciemat.es/data/neutrinos/DUNE/SOLAR/SENSITIVITY`

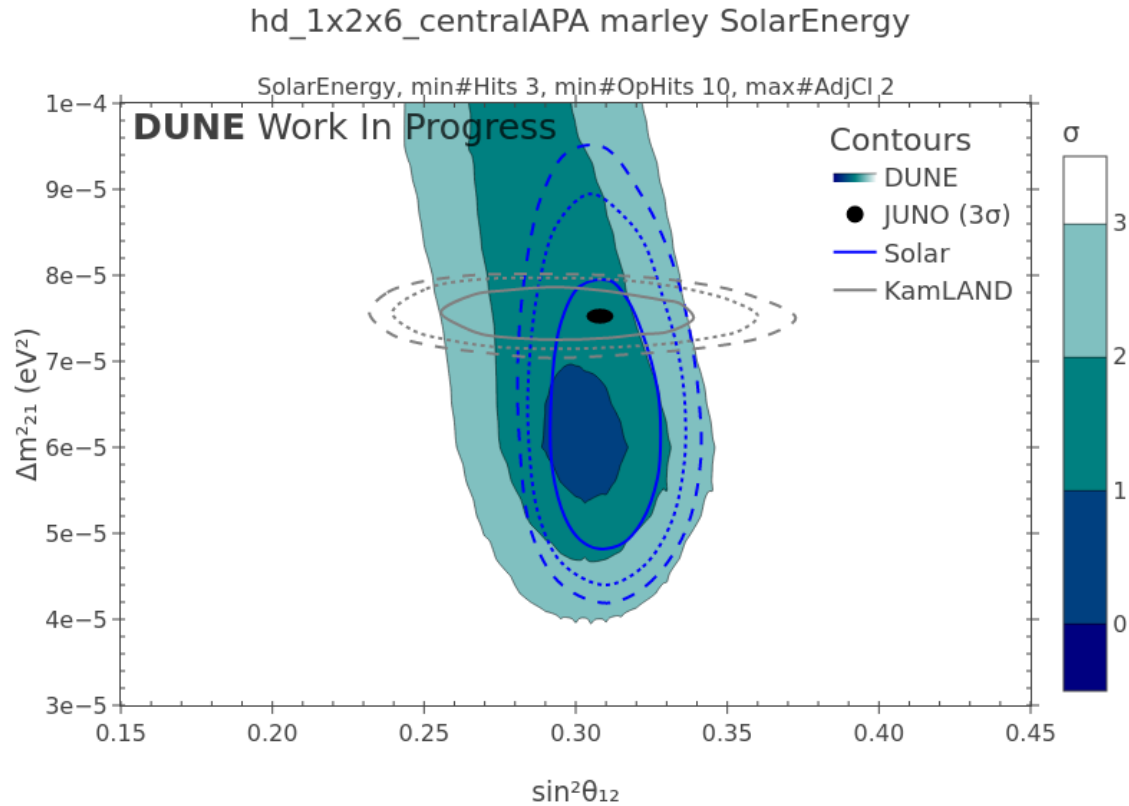
Sensitivity Fit Summary

- `src/analysis/14Sensitivity.py` builds fake observed maps as signal template at each oscillation point plus the corresponding background template, then fits that map against the solar and reactor reference templates with free signal and background normalizations.
- The fit in `lib/lib_root.py` minimizes a Poisson deviance-like objective, not a generic least-squares chi-square.
- For observed count o_i and expected count e_i , the per-bin contribution is $2(e_i - o_i + o_i \log(o_i/e_i))$ for $o_i > 0$ and $2e_i$ for $o_i = 0$, plus quadratic penalty terms on the fitted signal and background normalization shifts.
- The saved grid values are the minimized fit objective returned by `Sensitivity_Fitter.Fit` ; contour labels may display $\sqrt{\chi^2}$ as a visualization proxy, but the workflow fundamentally stores the minimized deviance / chi-square-like statistic itself.

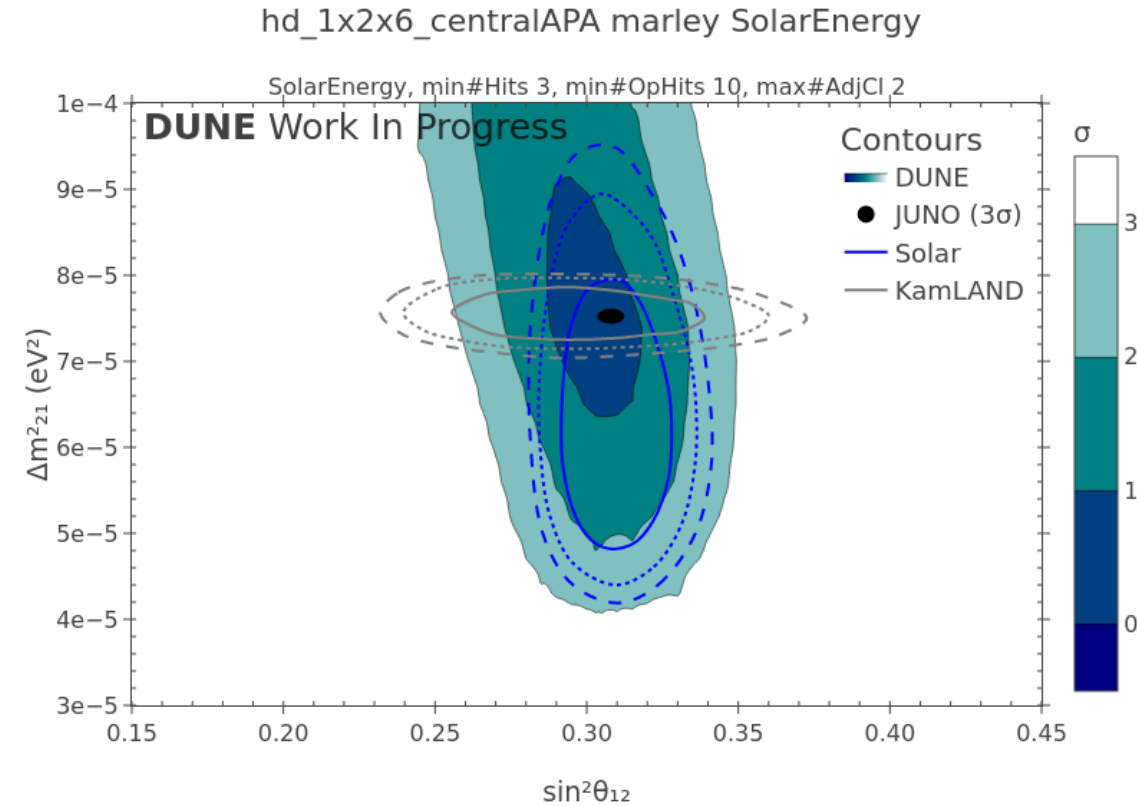
Main Result: Contour Grids (sin12)

HD Central

Solar Contour (sin12)

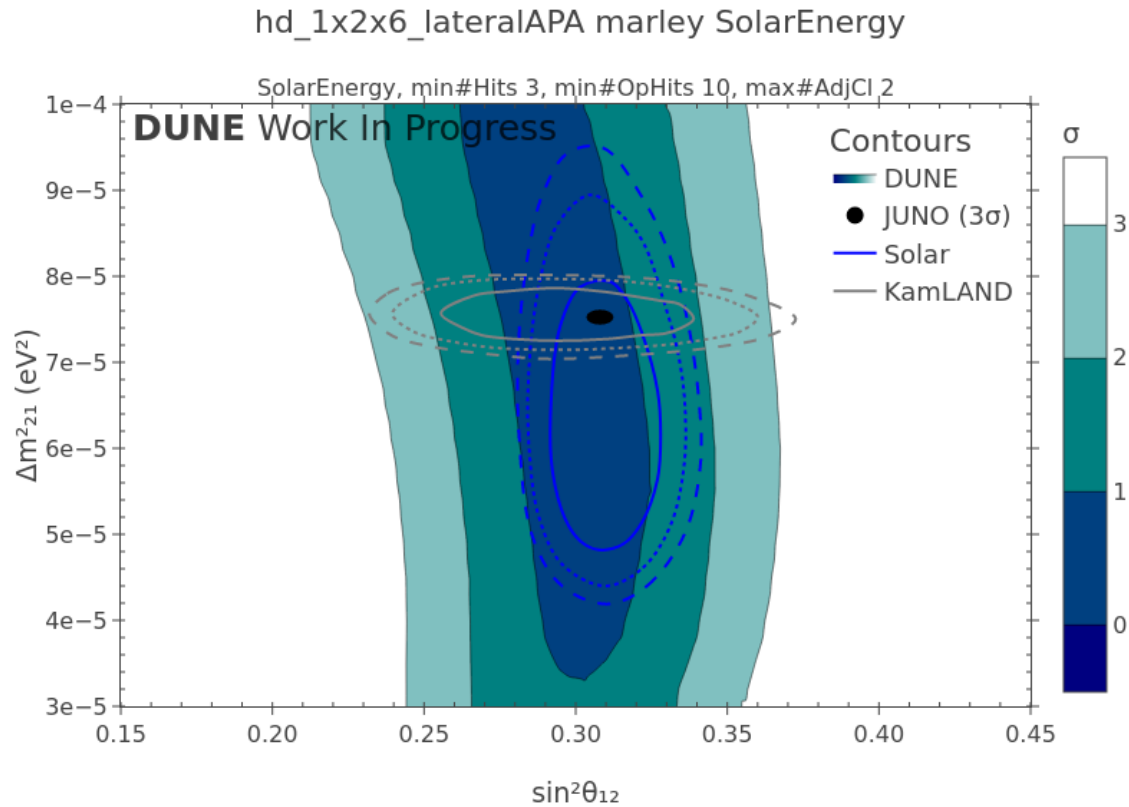


Reactor Contour (sin12)

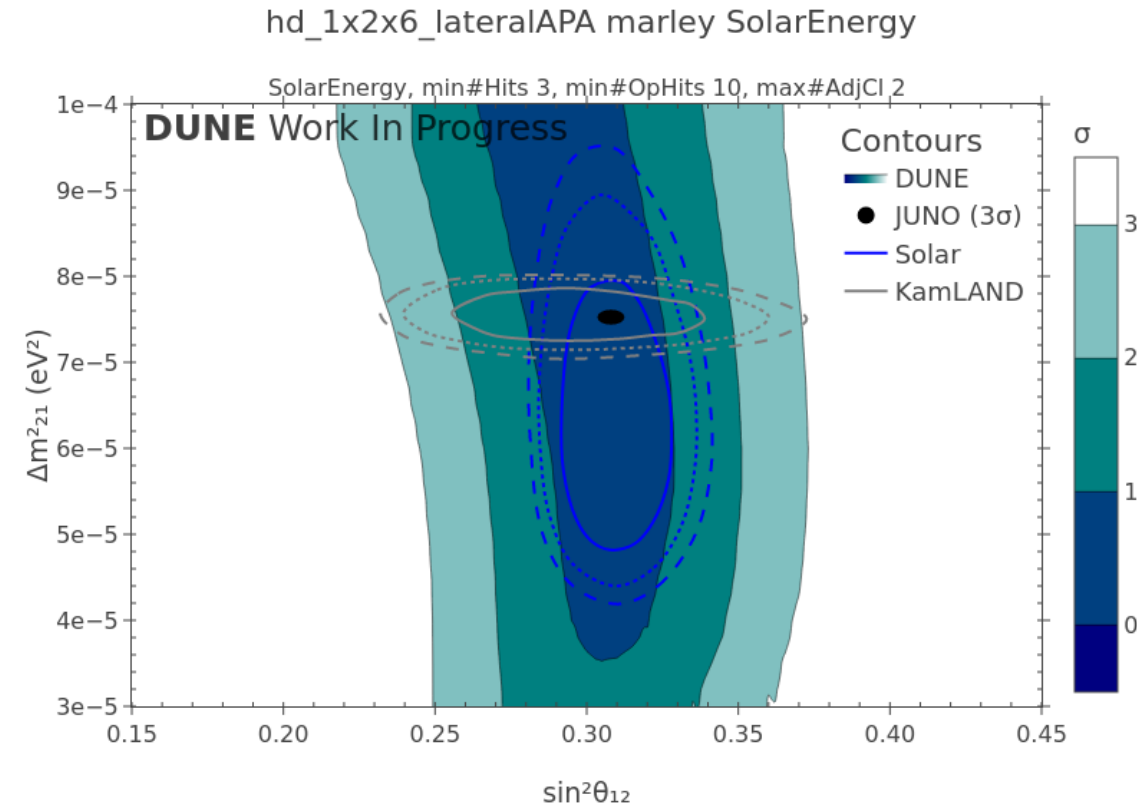


HD Lateral

Solar Contour (sin12)

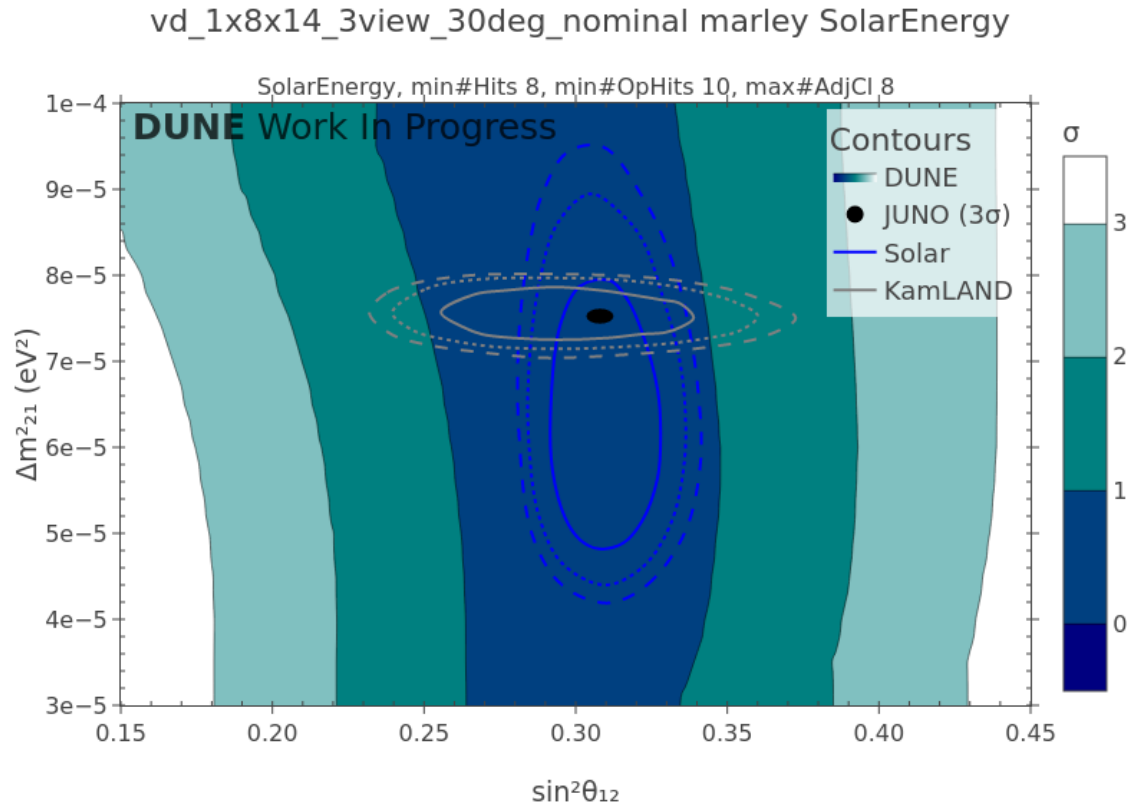


Reactor Contour (sin12)

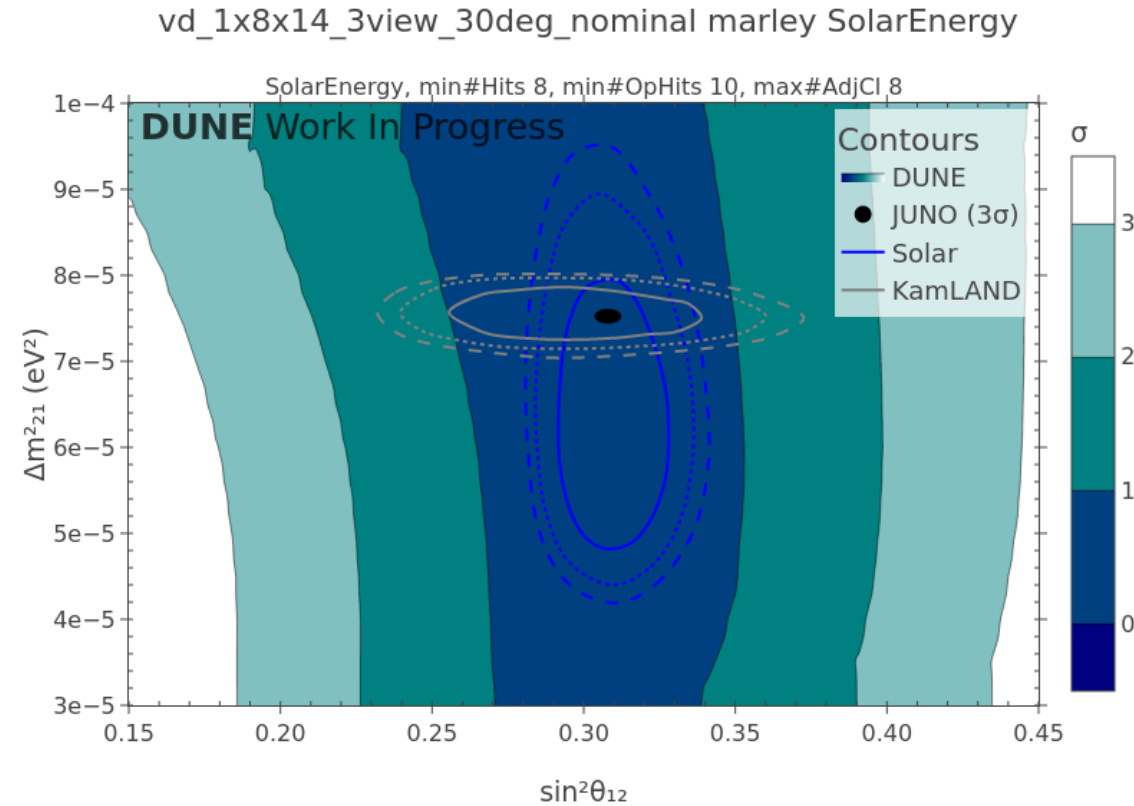


VD Top

Solar Contour (sin12)

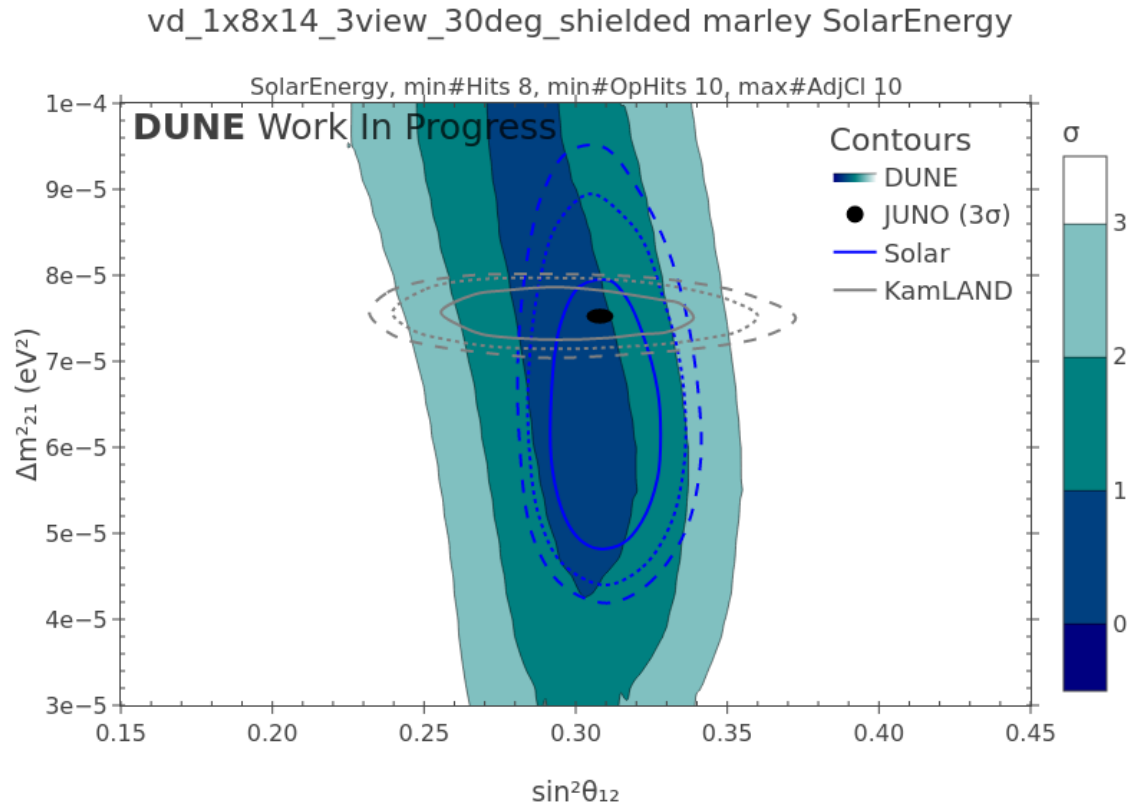


Reactor Contour (sin12)

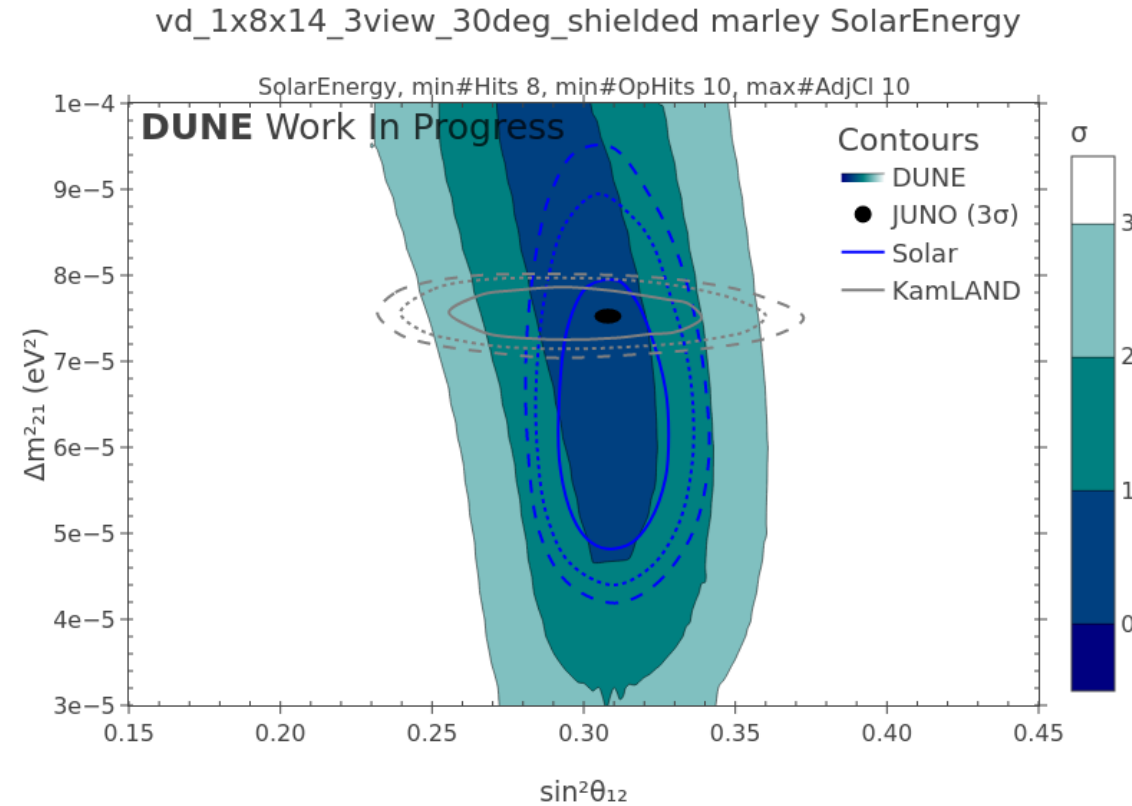


VD Bottom Shielded

Solar Contour (sin12)



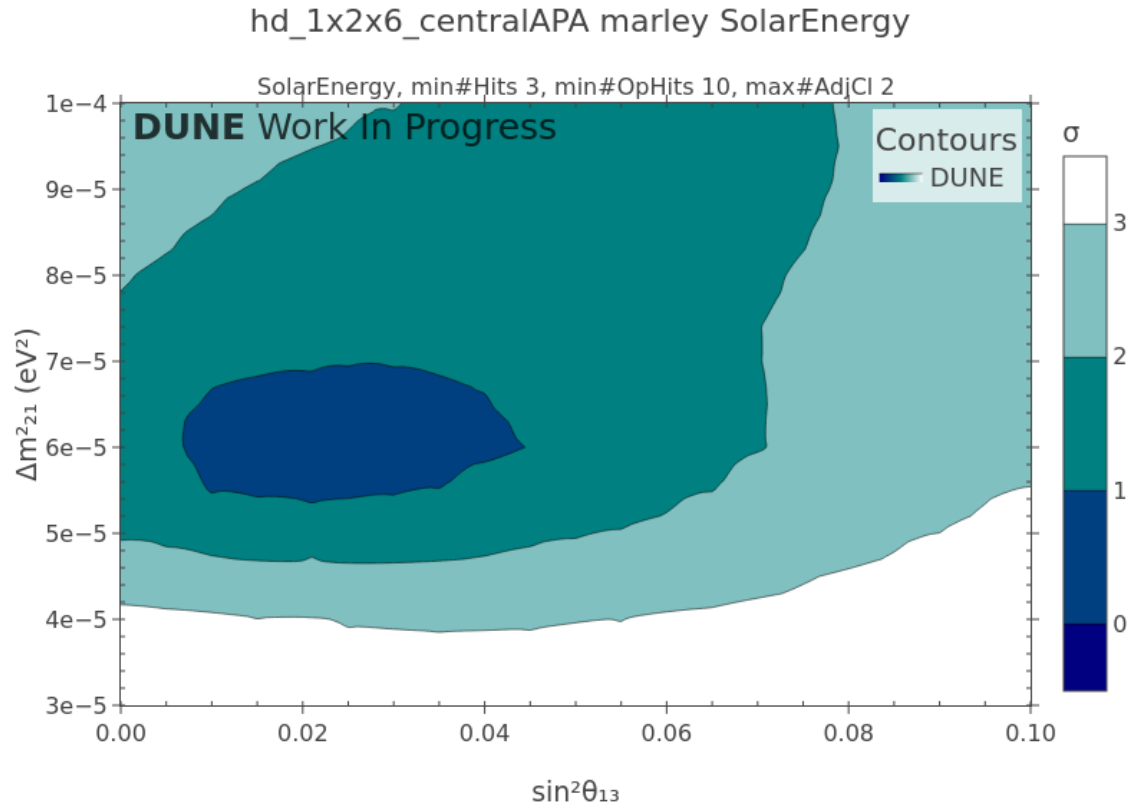
Reactor Contour (sin12)



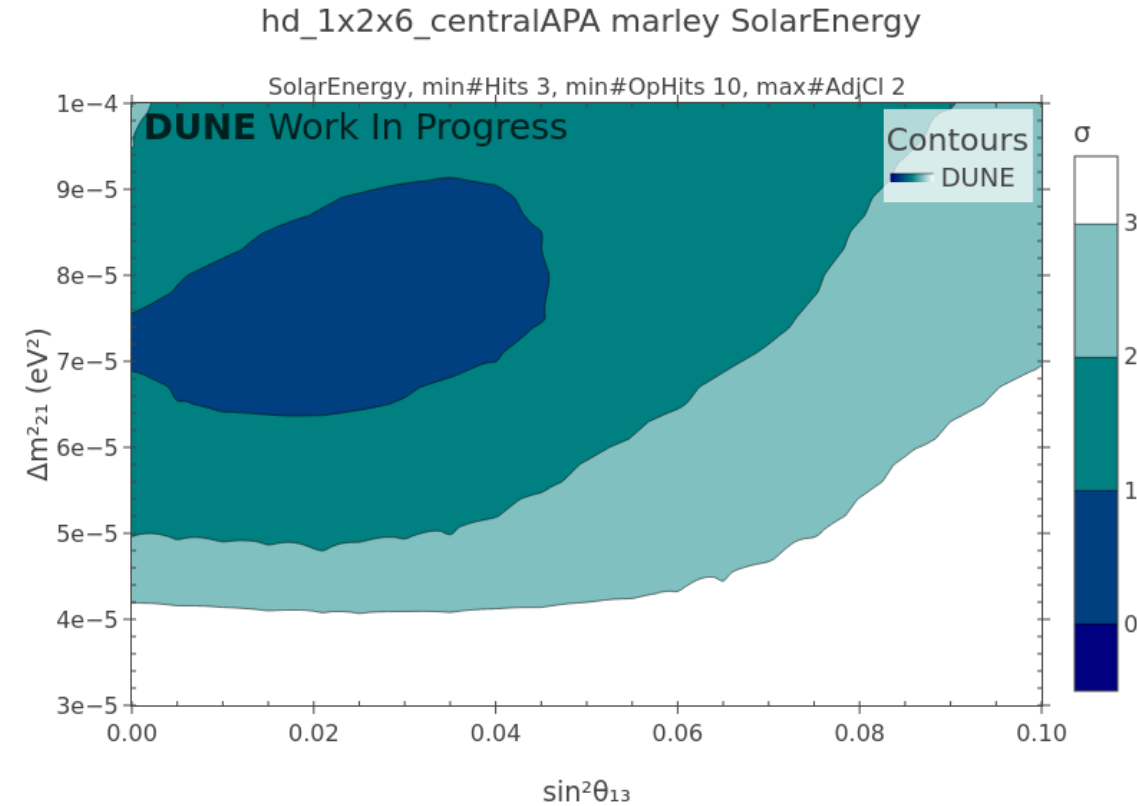
Contour Grids (sin13)

HD Central

Solar Contour (sin13)

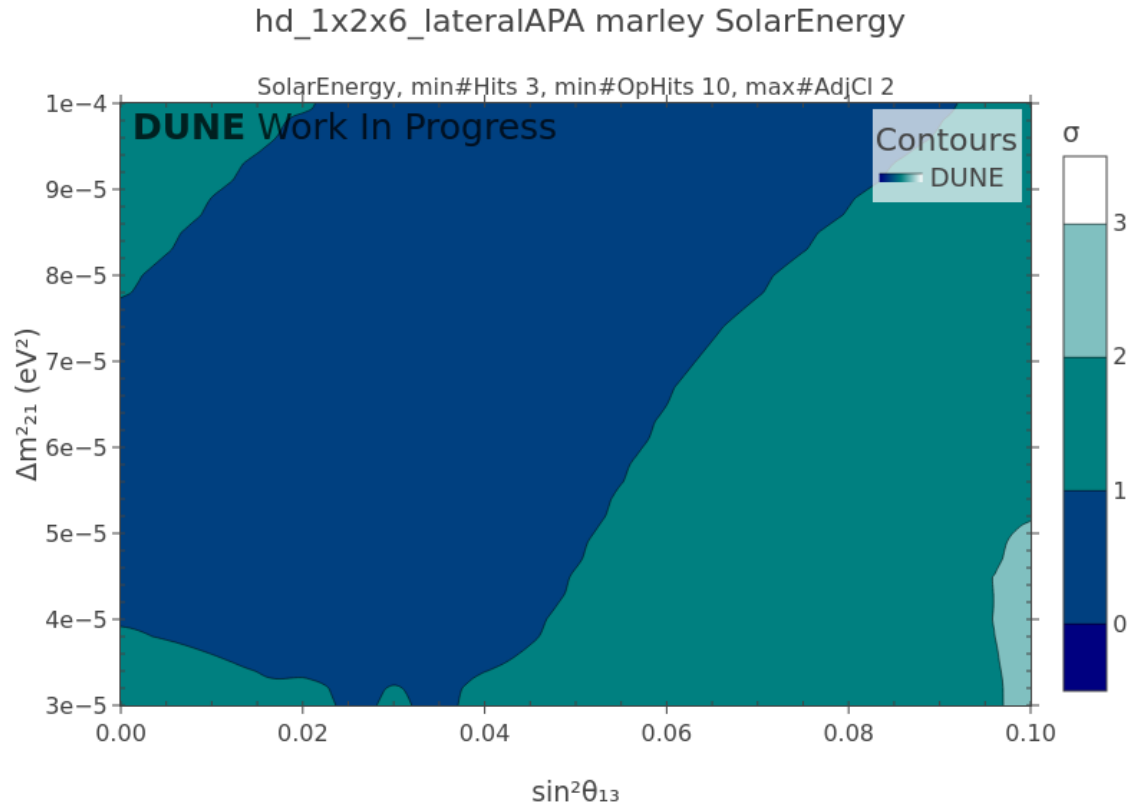


Reactor Contour (sin13)

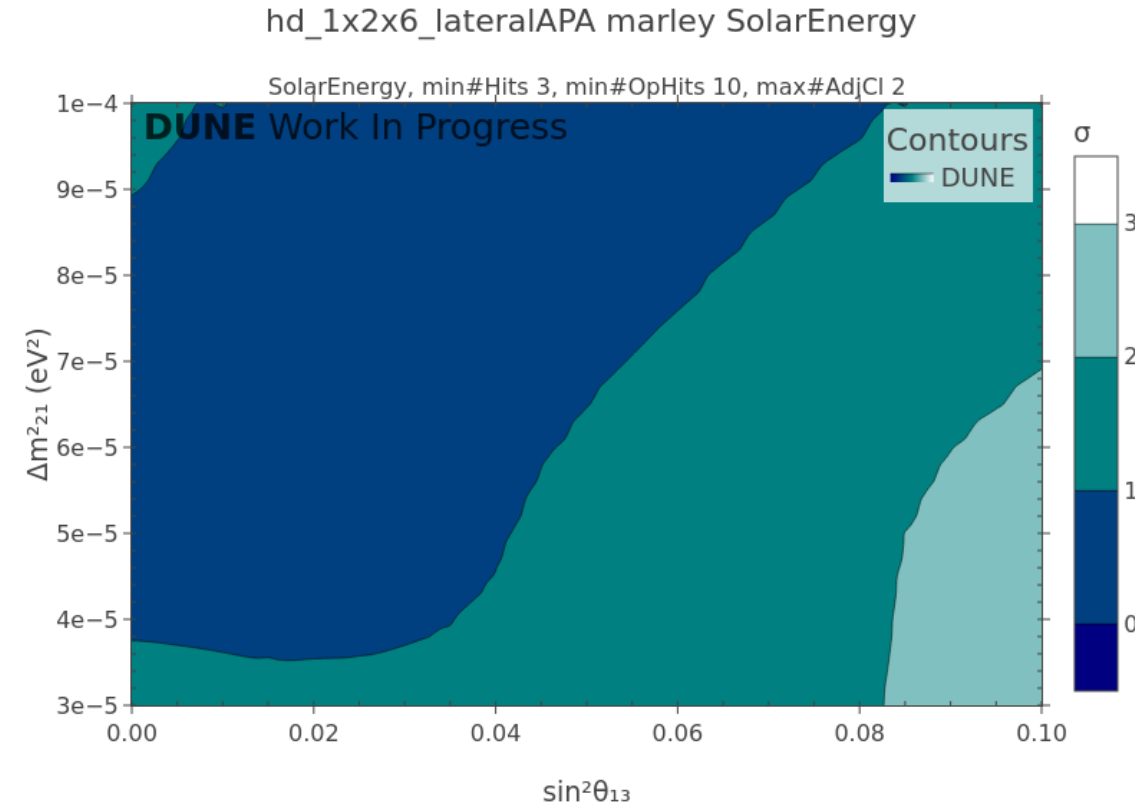


HD Lateral

Solar Contour (sin13)

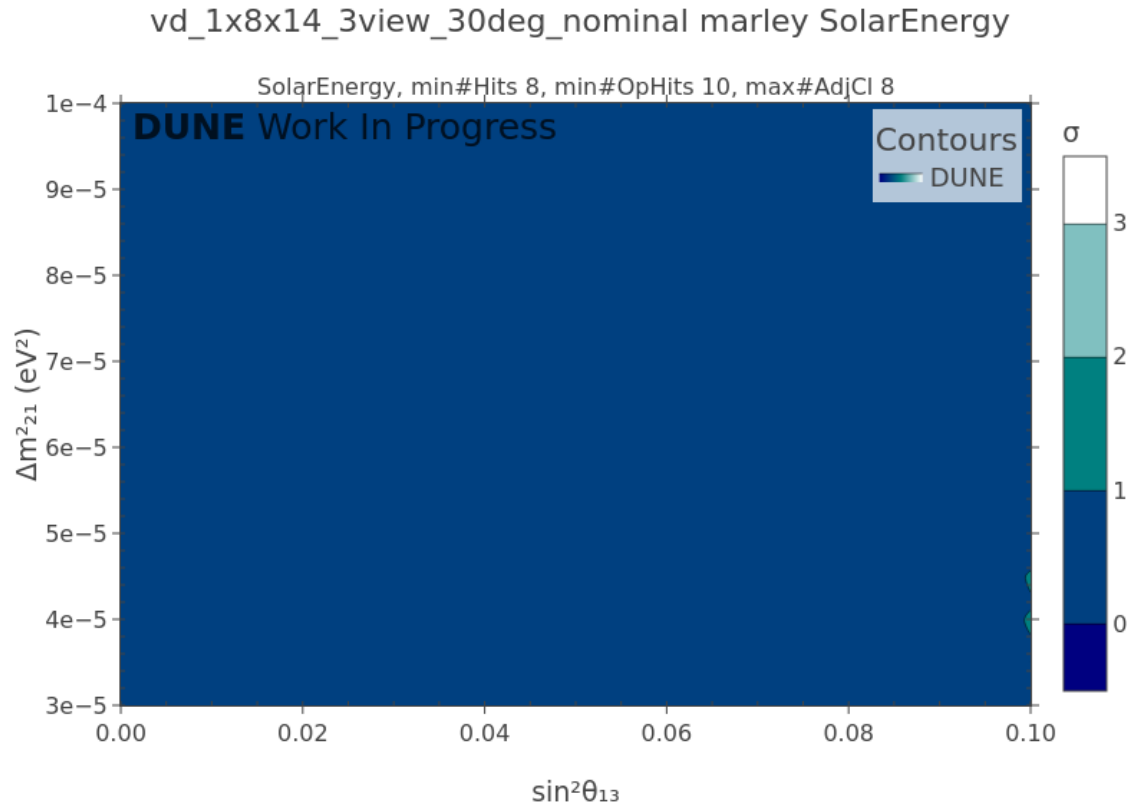


Reactor Contour (sin13)

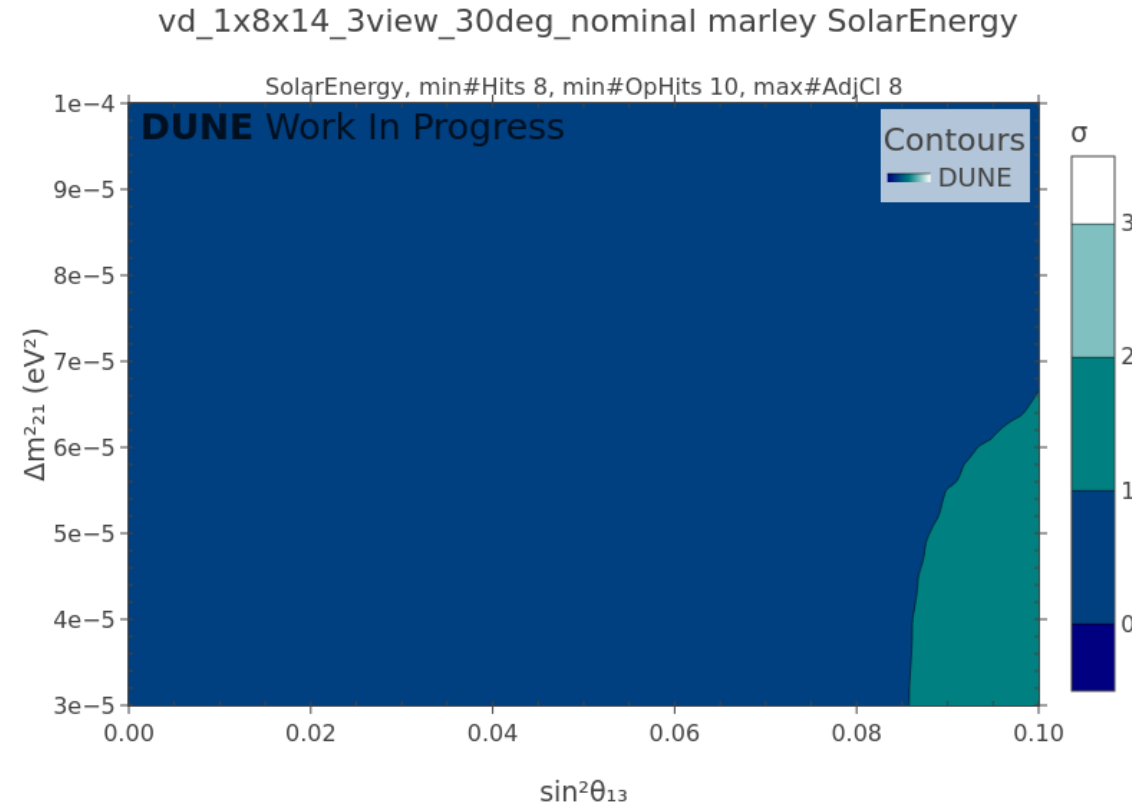


VD Top

Solar Contour (sin13)

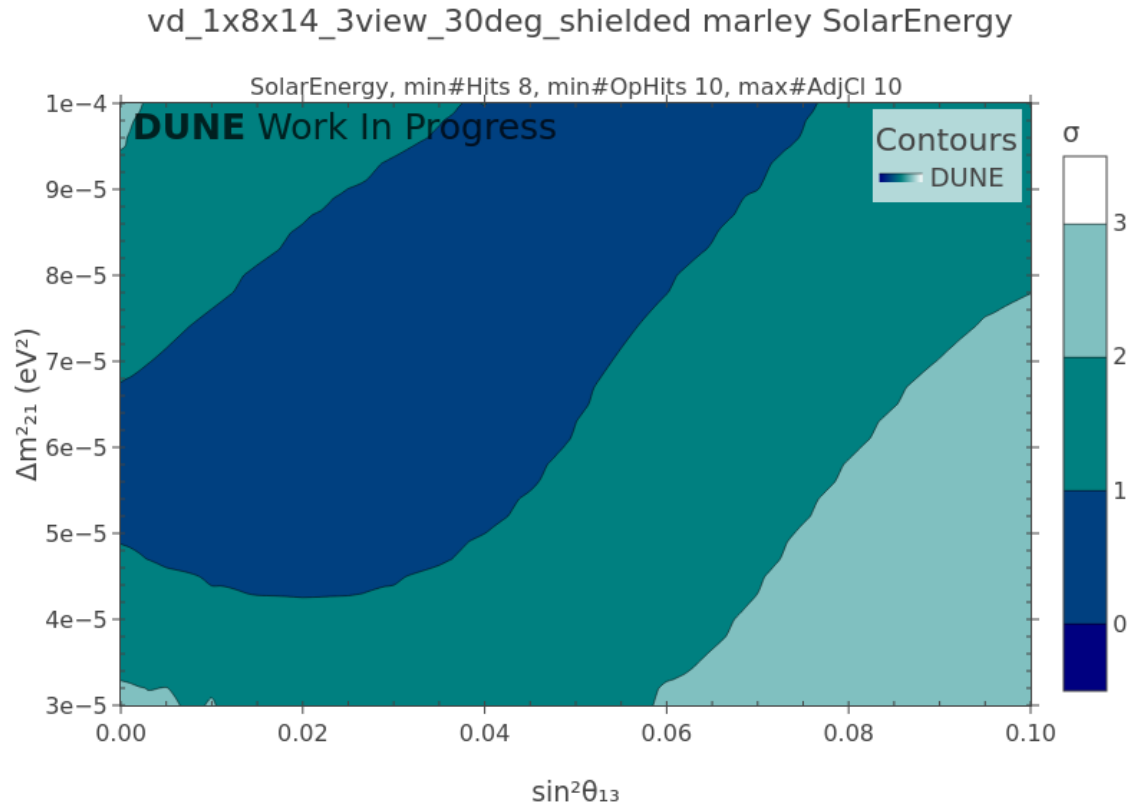


Reactor Contour (sin13)

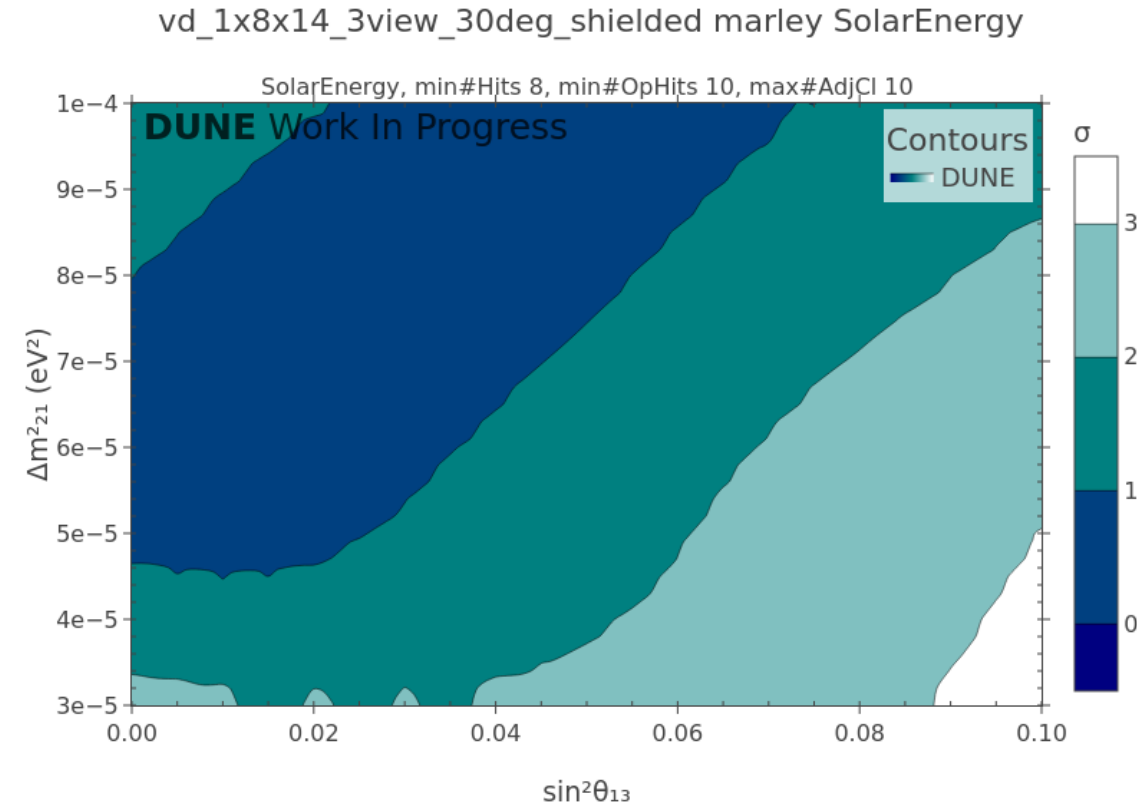


VD Bottom Shielded

Solar Contour (sin13)



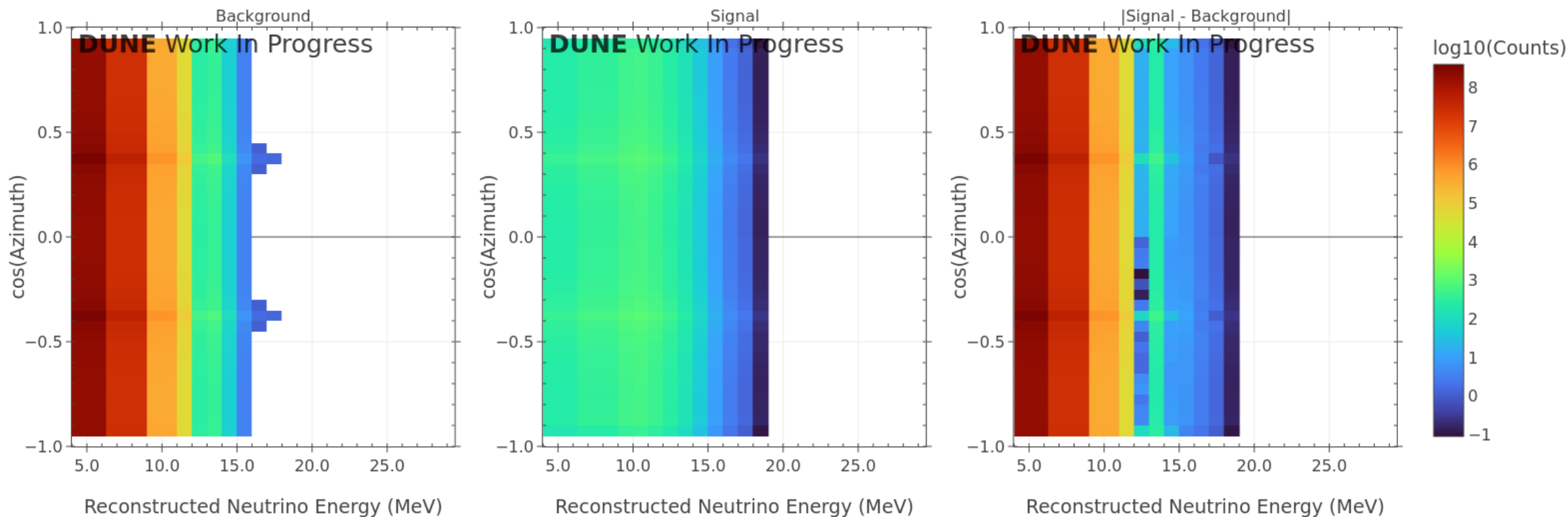
Reactor Contour (sin13)



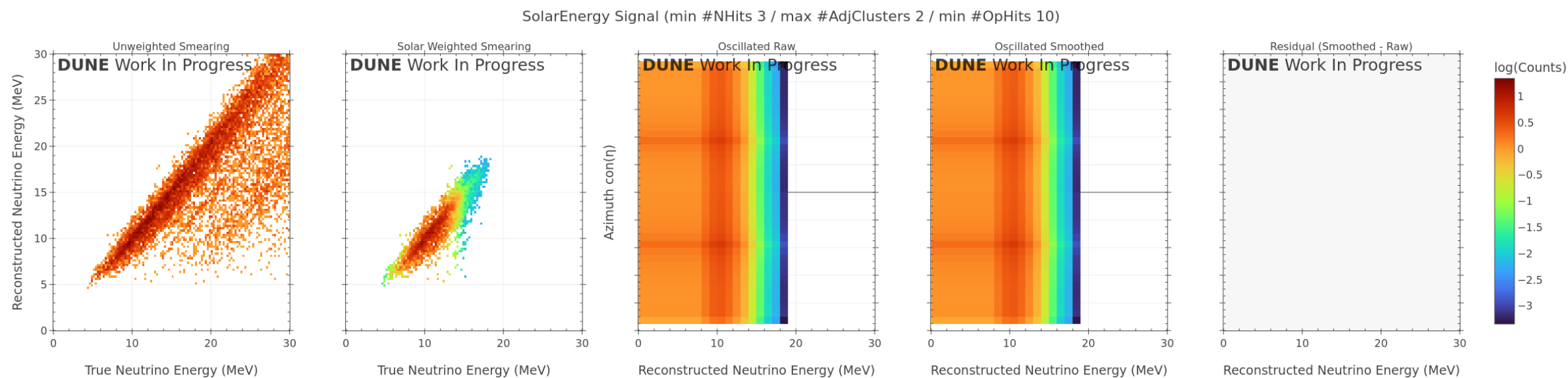
Template Building

HD Central Background Template

Sensitivity Templates hd_1x2x6_centralAPA marley SolarEnergy NHits3 OpHits10 AdjCI2

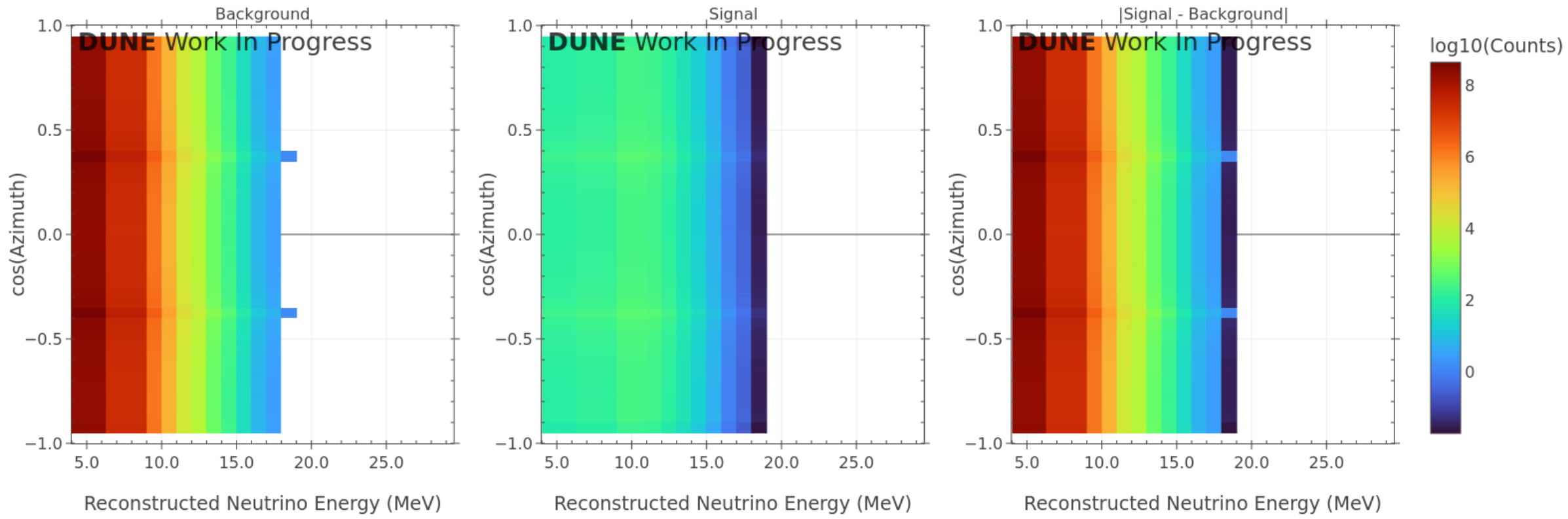


HD Central Signal Template

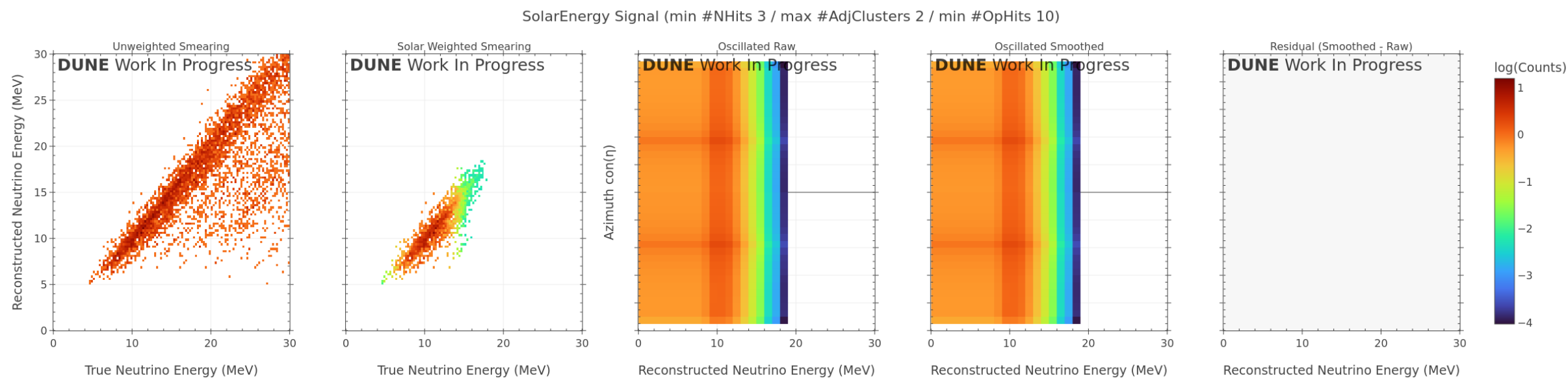


HD Lateral Background Template

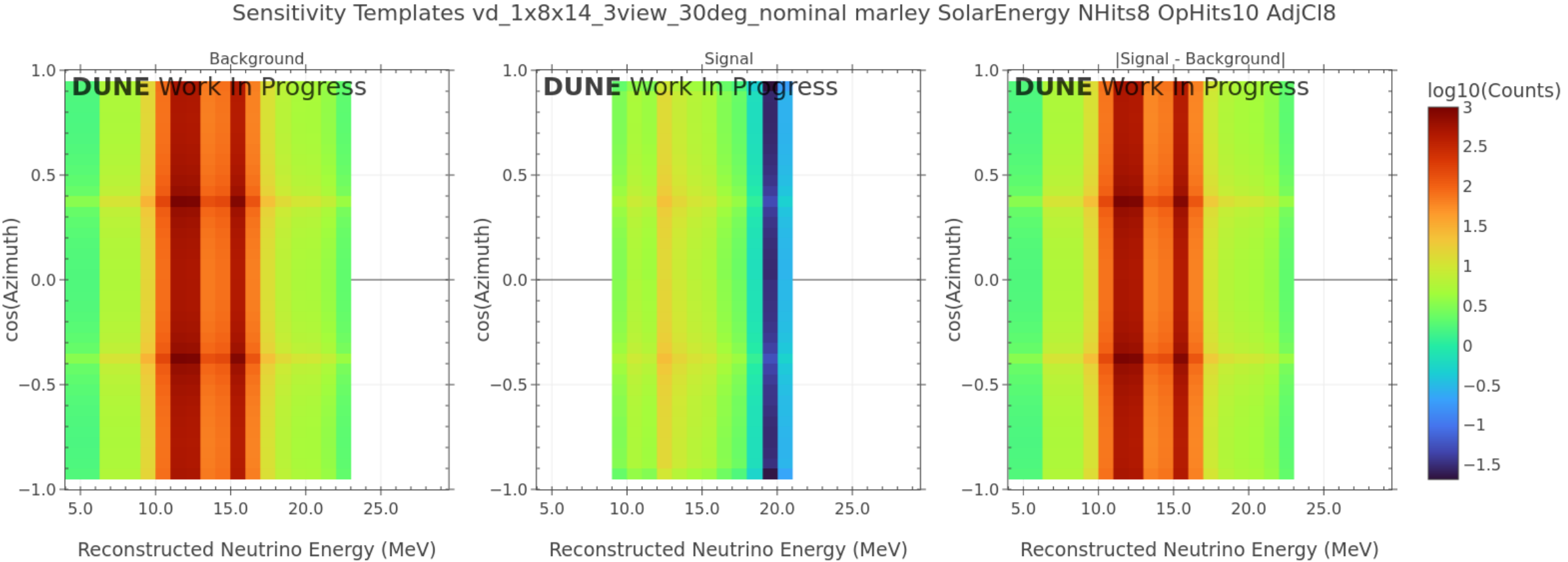
Sensitivity Templates hd_1x2x6_lateralAPA marley SolarEnergy NHits3 OpHits10 AdjCI2



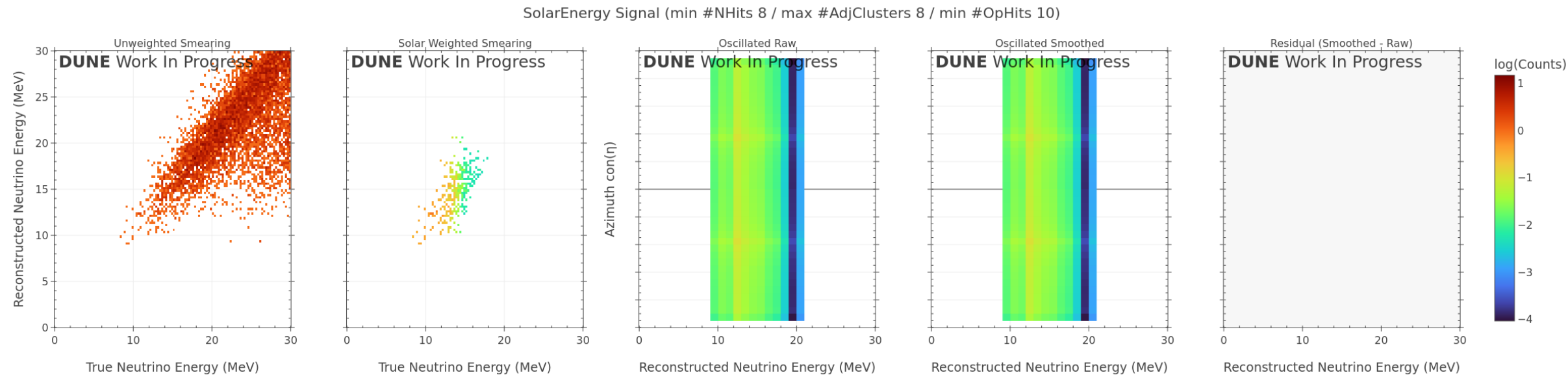
HD Lateral Signal Template



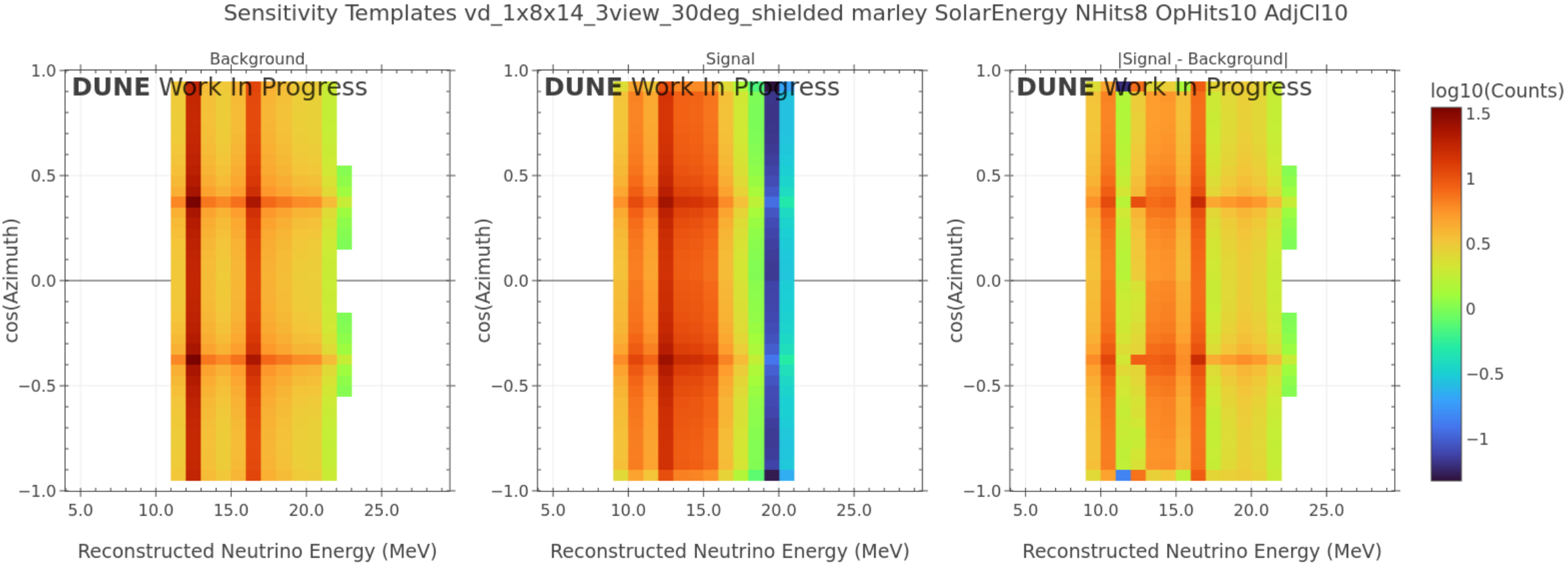
VD Top Background Template



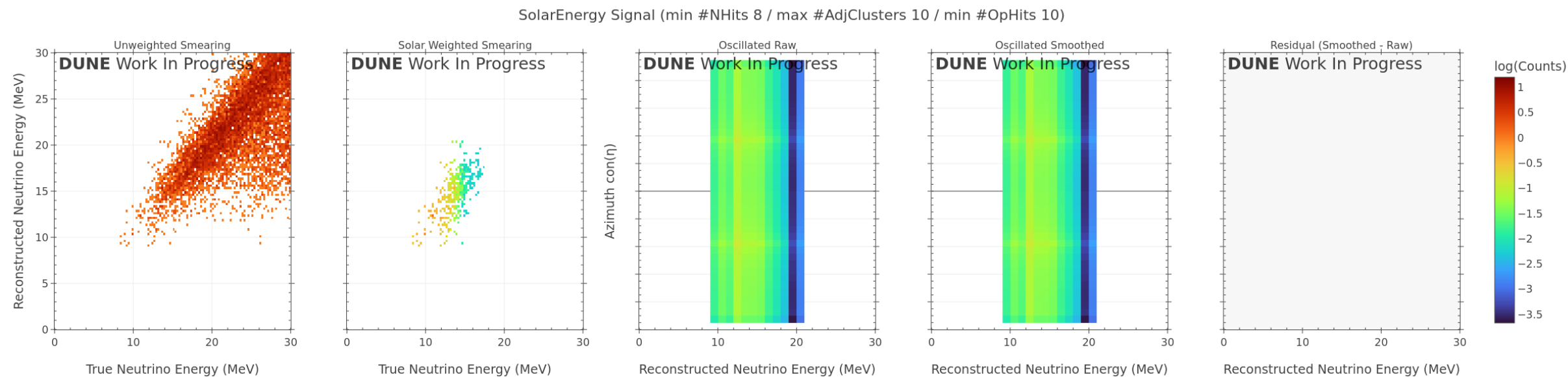
VD Top Signal Template



VD Bottom Shielded Background Template



VD Bottom Shielded Signal Template



Selected Sensitivity Cuts by Config

Config	NHits	OpHits	AdjCl	Signal Unc. (%)	Bkg Unc. (%)
HD Central	3	10	2	4	2
HD Lateral	3	10	2	4	2
VD Top	8	10	8	4	2
VD Bottom Shielded	8	10	10	4	2

Coverage and Notes

- Configs with selected sin12 solar contour plots:
- truncated: 4
- Cut table values are parsed from selected result filenames when available.
- Re-run script to refresh this folder after each workflow run:
- `/usr/bin/python3 scripts/generate_sensitivity_presentation.py --folder truncated`